

HACPO. Optimized Specifcation

This file refines the original HACPO (Hybrid Adaptive Contrastive Policy Optimization) objective. we 1) unify notation, 2) make clipping/PPO mechanics explicit, 3) define the contrastive term and weights, 4) give practical hyperparameter suggestions and 5) provide implementation notes and pseudocode.

Notation (unified).

- θ . policy parameters for $\pi_\theta(y \mid x)$.
- π_{ref} . reference / behavior policy (fixed when computing ratios).
- $r_i(\theta) = \frac{\pi_\theta(y_i \mid x_i)}{\pi_{\text{ref}}(y_i \mid x_i)}$ importance / policy ratio for supervised sample (x_i, y_i) .
- $r_j^{(k)}(\theta) = \frac{\pi_\theta(y_{j,k} \mid x_j)}{\pi_{\text{ref}}(y_{j,k} \mid x_j)}$ ratio for the k -th contrastive (negative) sample associated with context x_j .
- $\hat{A}_i$. estimated (possibly generalized) advantage for sample i . Use the same notation for all advantages for clarity.
- $w_{j,k} \geq 0$. weight for contrastive sample $y_{j,k}$; we require $\sum_k w_{j,k} = 1$ for each context j .
- Clip interval parameter $\epsilon > 0$; define $\text{clip}_\epsilon(r) = \text{clip}(r, 1 - \epsilon, 1 + \epsilon)$.
- L_{CPO} . contrastive policy optimization loss (defined below).
- $D_{\text{KL}}(\pi_{\text{ref}} \parallel \pi_\theta)$. KL divergence used as a constraint/penalty. (See note about direction and sign.)

Objective

We present the HACPO objective in a form consistent with PPO-style clipping and with an explicit contrastive term.

$$\begin{aligned}
 J_{\text{HACPO}}(\theta) = \mathbb{E}_{(x_i, y_i) \sim \mathcal{D}} & \left[\underbrace{\alpha_i \cdot \min(r_i(\theta) \hat{A}_i, \text{clip}_\epsilon(r_i(\theta)) \hat{A}_i)}_{\text{supervised/PPO-clip term}} \right. \\
 & \left. + (1 - \alpha_i) \cdot \underbrace{\sum_{k=1}^K w_{i,k} \min(r_i^{(k)}(\theta) \hat{A}_i^{(k)}, \text{clip}_\epsilon(r_i^{(k)}(\theta)) \hat{A}_i^{(k)})}_{\text{contrastive importance-sampled PPO-clip}} \right] \quad (1) \\
 & + \gamma \mathbb{E}_{x_i \sim \mathcal{D}, \{y_{i,k}\} \sim A} [L_{\text{CPO}}(\theta; x_i, \{y_{i,k}\})] \\
 & - \beta_{\text{KL}} D_{\text{KL}}(\pi_{\text{ref}} \parallel \pi_\theta).
 \end{aligned}$$

Notes on signs and KL direction. Many RL formulations either maximize an objective (PPO) or minimize a loss. Above we write J_{HACPO} as an objective to maximize; the KL term is written as a penalty by subtracting $\beta_{\text{KL}} D_{\text{KL}}(\pi_{\text{ref}} \parallel \pi_\theta)$. If you implement using gradient descent on a loss, negate J_{HACPO} accordingly. We use $D_{\text{KL}}(\pi_{\text{ref}} \parallel \pi_\theta)$ so that the penalty encourages π_θ to stay close to the reference policy [?].

Contrastive loss. L_{CPO}

One practical choice for the contrastive policy loss (inspired by InfoNCE-style contrastive objectives, adapted to policies) is.

$$L_{\text{CPO}}(\theta; x, \{y_k\}) = -\log \frac{\exp(s_\theta(x, y^*)/\tau_c)}{\sum_k \exp(s_\theta(x, y_k)/\tau_c)} \quad (2)$$

